

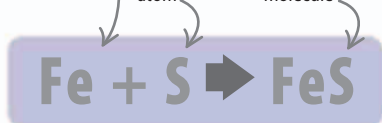
Equations

Chemists use equations to represent what is happening during chemical reactions. The formula of each reactant is written on the left-hand side, and those of the products are shown on the right. An arrow indicates the direction in which the reaction occurs.

Chemical symbols

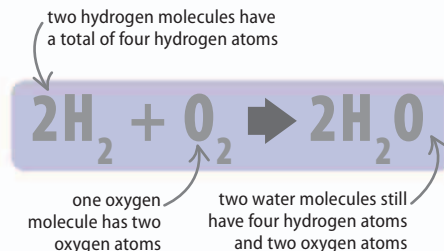
Instead of using the elements' names, their chemical symbols are shown.

one iron atom one sulfur atom one iron sulfide molecule



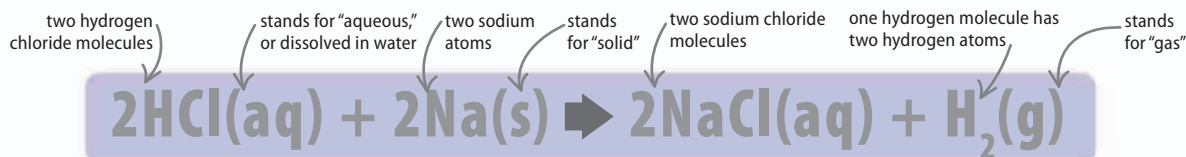
Balanced equations

The number of atoms in the reactants is the same as the number of atoms in the products.



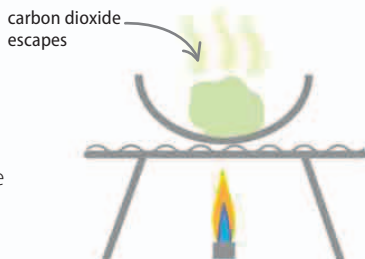
Reaction conditions

The equation can also contain other information about the reaction, such as the state of the reactants and products.



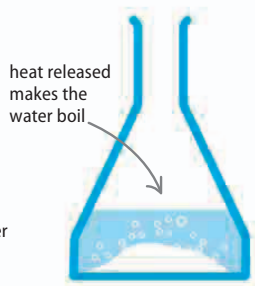
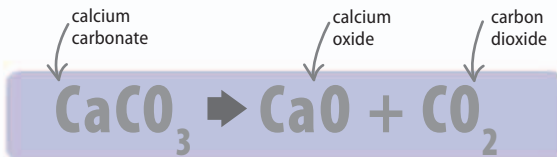
Types of reaction

There are three main types of chemical reactions. In a decomposition reaction, one complex product breaks apart into two (or perhaps more) simple products. In a synthesis reaction, two or more simple reactants join together to form a single, more complicated product. In displacement reactions, atoms or ions of one type swap places with those of another, forming new compounds.



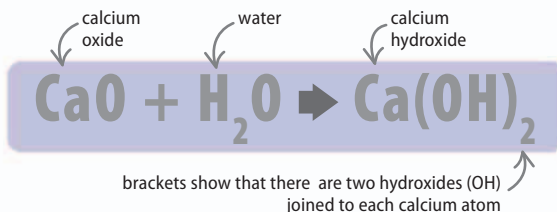
Decomposition reaction

Calcium carbonate (CaCO_3) decomposes into calcium oxide (CaO) and carbon dioxide (CO_2) when heated.



Synthesis reaction

Calcium oxide (CaO) powder and water (H_2O) combine in a synthesis reaction to form calcium hydroxide ($\text{Ca}(\text{OH})_2$), which dissolves in the remaining water.



Displacement reaction

When heated gently, carbon dioxide (CO_2) displaces the hydroxide in calcium hydroxide ($\text{Ca}(\text{OH})_2$) to make calcium carbonate (CaCO_3) and water (H_2O).

